



Ahmedabad
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GROUP 17

Probabilistic Adaptive Traffic Signal Control in Intelligent Transport Systems (ITS)

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Project Context and Motivation

Probabilistic Adaptive Traffic Signal Control in Intelligent Transport Systems

Problem

Old traffic signals use 'fixed-timers' based on historical averages. They can't see what's actually happening on the road.

? Why Uncertainty Exists

We can't predict exactly when a car will arrive $N(k)$. Every driver reacts differently. Traffic demand changes constantly — rush hour vs. midnight.

The Result

Fixed signals waste green time when roads are empty and cause massive jams during unexpected surges.

Our Goal

Use probability to make signals 'think' and to make proactive, real-time traffic decisions, not reactive fixes.

Objective:

Build a model that minimizes queue length and vehicle delay by calculating the probability of future traffic states and selecting the optimal signal phase.

Scopes:

- Standard 4-way intersection
- 8 directions
- 256-state Markov Process (2^8 binary congestion states)

Estimated:

- Calculate how fast cars are arriving using sensors
- Estimated via exponential smoothing of detector counts
- Updated every $\Delta t = 3$ second cycle — real-time adaptation

The Decision:

- Choose Green or Red for the next 3 seconds
- Policy π^* maps each state $S(k)$ to the optimal action
- Minimizes expected future queue accumulation across all movements

System Pipeline: How It Works



Data Flow

- Stochastic Input (Arrivals) → State Update (Queue Evolution)
- Probabilistic Decision (Markov Policy) → Output (Discharged Vehicles)
- All 8 movements updated simultaneously each $\Delta t = 3$ sec cycle

Why This Works

- Loop detectors give real-time $Q(k)$ — not historical averages
- Poisson model converts raw counts to probability distributions
- Bellman equation guarantees globally optimal phase selection

 Closed-Loop Logic: What we decide NOW affects how many cars will be there in the NEXT step. The system continuously learns and adapts.

Sources of uncertainty: Why It's Hard



Random Arrivals

- Even knowing 10 cars arrive per minute on average — they don't arrive at perfect intervals
- This is natural, irreducible randomness captured by the Poisson process
- No deterministic predictor can overcome this fundamental stochasticity



Environmental Changes

- Rain, accidents, or events can shift traffic flow by 15–20%
- Demand pattern changes from peak to off-peak unpredictably
- λ itself must be estimated — it is not a fixed constant



Driver Behavior

- Different reaction times per driver change departure headways
- Mix of trucks and passenger cars makes clearing speed unpredictable
- Creates variability in q_{out} even when signal is Green



Binary Limits

- Traffic is labelled as just Congested or Not Congested
- This loses fine-grained detail but keeps computation fast
- Accepted trade-off: queue of 11 treated same as queue of 50 if $T = 10$

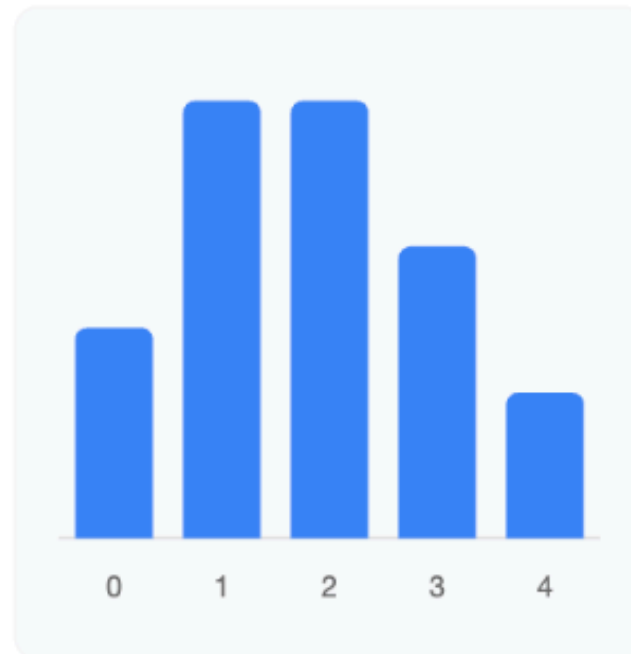
Key Random Variables and Their Roles

STOCHASTIC INPUT

Arrivals N_k

$$N_k \sim \text{Poisson}(\lambda \cdot \Delta t)$$

Number of vehicles arriving in interval Δt . Follows a Poisson distribution characterized by mean arrival rate λ .

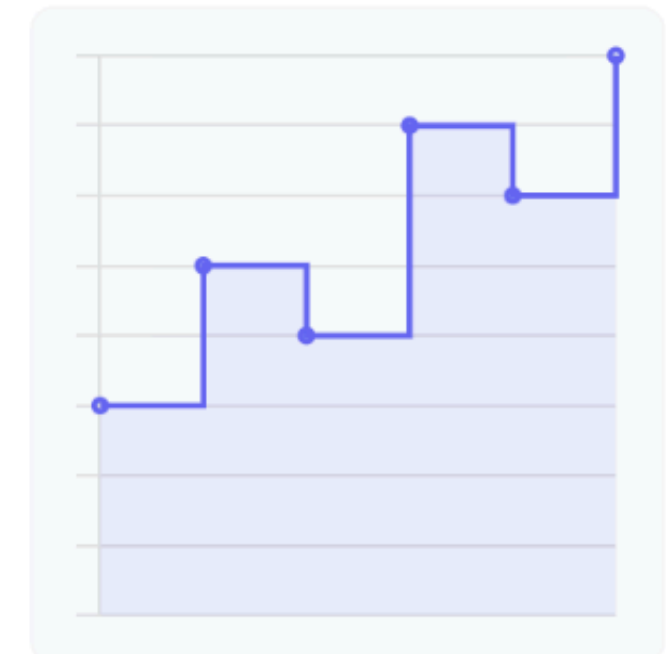


SYSTEM STATE

Queue Q_k

$$Q_{k+1} = \max(0, Q_k + N_k - q_{out})$$

Modeled as a **Reflected Random Walk**. The queue grows with arrivals and shrinks with departures, never dropping below zero.



BINARY CLASSIFIER

Congestion S_k

$$S_k = \mathbb{1}(Q_k \geq T)$$

A simplified binary state (0 or 1). Active when queue length exceeds threshold T . Helps reduce state space complexity.

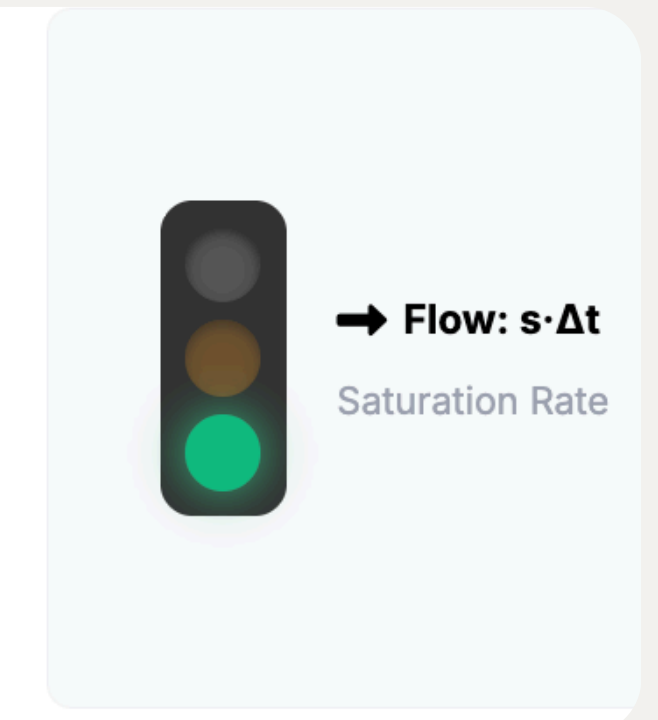


CONTROLLED OUTPUT

Departures q_{out}

$$q_{out} = \min(s \cdot \Delta t, Q_k) \cdot \mathbb{1}(\text{Green})$$

Depends on control $u(k)$. Zero if Red. If Green, limited by saturation flow s or available queue.



How the System Infers

The Uncertainty

- Because arrivals are random, the queue isn't a fixed number; it's a probability cloud.

How it can be done

- We calculate the chance of a jam before it happens:
Eg: If the light is red, and for the particular lane there is an 80% chance the queue will exceed the threshold in 3 seconds.

Justification

- Therefore, the system bets against congestion. It chooses the signal phase that results in the lowest expected queue length.

Stastical Foundations

Mean:

- It solves how heavy the traffic load is. We use it to check if the intersection has enough physical capacity.

Median:

- It solves what is the typical traffic volume and tells us if our signal plan is normal for most of the hour.

Variance:

- It solves how unpredictable the traffic is. Most informative metric, as it tells high variance means traffic is deadly.

Inequalities to be used

Jensen Inequality

- Shows us how the average data is a trap. We can use it to prove that chaotic traffic mathematically creates more delay than averages predict, justifying better adaptive solution.

Markov Inequality

- Gives us a worst case safety bound using only the average queue length. We use this to set safe queue thresholds.

Chebysev Inequality

- Uses variance to predict extreme spikes. We can use it to identify when the road reaches its maximum capacity and when the queue length is at its maximum.

Chernoff Bound

- It solves peak hour surge planning for rare events. We can use it as the tool to trigger emergency green lights before the intersection reaches a total chaos.

Current Limitations and Conceptual Gaps

- The threshold for deciding the congestion state is currently a fixed number (T); it should ideally change as traffic gets heavier.
- Our model only sees if its busy or not busy. We need to add a medium state for more detail.
- As cars often arrive in groups due to nearby signals, which can disrupt the Poisson model.

Planned Refinements and Role Coordination

Next Milestone Evolution:

- Moving from equations to a functional simulation.
- Testing model on rush hour traffic.

Tasks Ahead:

- Model coding and extended simulations.
- Checks if the math holds up with real life situations.

Role Coordination:

- Reviewer validates assumptions and logic.
- GitHub manager handles version control and ensuring simulation run efficiently.

Conclusion

Problem with Fixed Timers

- We also know that fixed timers fail because they ignore the average time during high traffic hardly works, and can be provided with Jensen Inequality
- Without a probabilistic model, a signal can never be truly efficient during peak hours.

The Probabilistic approach

- We know that traffic is never steady; it is a probabilistic process where variance is the main cause of delays.
- The model is designed to bet against congestion by calculating the odds of a jam and picking the signal phase with the lowest expected queue length.

Mathematical Logic & Next Steps

- We have selected math tools like Jensen's and Chebyshev's inequalities to help us predict traffic spikes and set safe limits during the simulation.
- With this mathematical blueprint complete, we can convert this theory to a working Python simulation in Milestone 3.